

Curiosity Guided Reinforcement Learning for Collaborative UAV and Quadrupe Robot Search

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Abstract—Collaborative heterogeneous robot systems can perform complex tasks efficiently and flexibly, enhancing the effectiveness of locating and navigating to targets. This article designs a heterogeneous robot system consisting of unmanned aerial vehicle (UAV) and quadrupe robot for performing search and rescue tasks in unknown and complex environments. This system can use a multi-stage reinforcement learning framework and deep reinforcement learning strategy algorithm to achieve guiding robots by UAV to search and find targets in mines. Among them, quadrupe robot is capable of navigating uneven terrain and adapting to dynamic environments, making it more capable of overcoming considerable number of obstacles than traditional tracked or wheeled robots. By establishing a stable model, the robot is equipped with the foundation necessary to perform more complex tasks, such as navigating cluttered environments or overcoming obstacles. The development of such fundamental behaviors is critical for future applications in search and rescue, exploration, and industrial tasks, where adaptability and precision are paramount. Experimental results in simulation environments demonstrate the effectiveness of the training model, highlighting its ability to grow and improve.

Keywords—Heterogeneous intelligent systems, Deep reinforcement learning, Multi-agent system, Target search

I. INTRODUCTION

In recent years, the development of legged robots such as quadrupe has attracted great attention because they can navigate challenging terrains and environments that wheeled or tracked robots find difficult to handle. In complex and unknown mining environments, quadrupe robots have several key advantages over unmanned ground vehicle (UGV). Firstly, quadrupe robots have high terrain adaptability and can cope with uneven or inclined mining terrain through different gait patterns, avoiding UGV from slipping or getting stuck in complex terrains such as gravel and potholes. In addition, quadrupe robots can flexibly adjust their center of gravity and support point positions to maintain balance in narrow spaces or unstable surfaces, making them more passable in mines with irregular terrain and limited space. Secondly, quadrupe robots have excellent obstacle crossing capabilities, able to overcome uneven obstacles or in larger crevices, which are difficult to achieve with traditional UGV. Overall, the flexibility, stability, and adaptability of quadrupe robots give them significant advantages in complex and unknown environments such as mines.

One of the long-term goals of developing robots is to develop autonomous, agile, and safe products that can operate in complex environments [1]. For example, Boston Dynamics' Atlas can perform various agile actions through hydraulic

actuators [2]. However, achieving stable and efficient walking behavior in quadrupe robots remains a complex task due to their high degree of freedom, dynamic balance, and continuous adaptation to different surfaces and obstacles.

Traditional control methods for legged locomotion often rely on meticulously hand-tuned parameters and predefined gait patterns. While these methods have achieved some success, they can be limited in terms of adaptability and scalability, especially when facing unpredictable environments. With the rise of artificial intelligence and reinforcement learning, machine learning-based methods have emerged as promising alternatives for training robots to learn locomotion behaviors autonomously through interaction with their environment.

As the name suggests, a heterogeneous robot system is a system that integrates multiple robots. The advantage of such a system is that it can efficiently coordinate different types of robots, integrate their abilities into one system to achieve division of labor planning and complete common tasks [3]. The system studied in this paper consists of a drone and a quadrupe robotic dog. The motion of the drone can be regarded as simple rigid body motion, which is easy to train. The training of quadrupe robotic dogs is a huge and challenging task. Robot dogs have many degrees of freedom and a large number of joints, so the dimensions of state space and action space are high, making it difficult to maintain correct posture and control their movements, which involves complex reward settings.

Currently, machine learning has been applied in multiple fields such as medicine [4], architecture [5], computer technology [6], and has achieved outstanding results. Among them, algorithms are a crucial part. Near end policy optimization (PPO) is based on an improvement of the policy gradient method, which has advanced performance and is easier to implement and adjust [7]. In order to use robotic dogs for training in more complex scenarios in the future, this article presents an attempt to train a quadrupe robotic dog (Spot) based on PPO in Unity [8]. The model of this Spot is based on an open-source project on GitHub that can be run manually or automatically on VirtualBox. And the scenario is developed from a scene related to rescue robots, aiming to conduct scalability research on this project. The first step involved in this paper is to first implant the Spot into the scene and conduct specialized training on its walking. The main contribution of this paper is to propose and validate a quadrupe robot motion control method that combines curiosity guided and goal driven strategies:

(1) In terms of posture stability, a reinforcement learning method based on the Intrinsic Curiosity Module (ICM) and PPO2 algorithm is adopted to enable quadruped robot to obtain exploration power through intrinsic rewards in sparse reward environments, effectively adjusting and optimizing their own posture to achieve stable motion posture.

(2) In terms of forward ability, a distance driven forward strategy has been designed to enable the robot to move autonomously over short distances based on its own position in inference testing.

By combining these two strategies, the feasibility of attitude control and navigation of robots in heterogeneous systems has been verified, laying the foundation for future collaborative tasks of robots in more complex environments and providing important references for related research.

II. RELATED WORKS

A. Heterogeneous robotic system

Heterogeneous robot systems have become a hot research topic in the field of robotics. Such systems are typically composed of multiple types of robots, covering three domains: land, sea, and air. A group of robots is typically able to complete tasks that a single robot cannot, as the diversity of robots increases robustness and flexibility, with each type of robot having different abilities [9].

Many researchers have successfully achieved the collaboration between unmanned aerial vehicle (UAV) and UGV, including UAV guidance and UGV tracking. In [10], a method for collaborative search and target discovery between aerial and ground robots was proposed. This method adopts a phased reinforcement learning framework and reinforcement learning, training the skills of two types of robots separately, and finally enabling them to work together. Among them, unmanned aerial vehicles equipped with laser radar lead ground vehicles forward and autonomously search for unknown target positions. The results demonstrate that the method is both efficient and effective, exhibiting robust performance across various environments. However, the obstacle crossing ability of UGV is limited, which may result in being trapped or unable to move forward, or increasing the distance of movement to adapt to mobility, making it difficult to obtain shorter travel routes, which may affect rescue time. In the work of this project, UGV will be replaced by quadruped robotic dog Spot to more efficiently complete more complex tasks.

B. Quadruped robot Spot

Quadruped robots represent a significant advancement in robotics, especially in challenging environments. Their design enables them to navigate complex terrains and perform tasks that typically exceed the capabilities of wheeled or tracked robots. One notable advantage is their exceptional adaptability to different terrains. Unlike wheeled or tracked robots, which can struggle with uneven surfaces, quadruped robots can adjust their gait and posture to maintain stability and traction. This adaptability enables them to traverse rocky landscapes, muddy fields, and steep slopes.

Compared with tracked robots and wheeled robots, legged robots only require discrete points instead of continuous motion when planning their motion paths, which allows them to adapt to more complex terrains [11]. Their four-legged design allows them to maintain mobility in the presence of

obstacles, significantly enhancing their operational range compared to wheeled and tracked vehicles.

Additionally, due to the length of the legs, robots can freely choose the contact points between their feet and the environment [12]. Quadruped robots offer superior stability and balance due to their four points of contact with the ground, distributing their weight effectively. This stability is particularly beneficial in hazardous mining environments, as they can move on unstable ground without tipping over. Their ability to maintain balance under different conditions gives them an advantage over wheeled and tracked robots, which may easily tip over.

In conclusion, quadruped robots offer numerous advantages over wheeled and tracked robots, particularly in complex and unpredictable environments. Their adaptability, obstacle negotiation capabilities, stability, versatile motion, and collaborative potential highlight their superiority in many scenarios. With the continuous advancement of technology, quadruped robots can play an increasingly important role in various fields.

C. The Current Status and Challenges of Applying Reinforcement Learning to Robot Motion Control

In recent years, due to the high computational requirements and the tendency to generate non optimal solutions, traditional motion planning algorithms often encounter problems in complex environments. Reinforcement learning (RL) can become a better alternative, adapting to dynamic environments in real-time and making optimal decisions [13].

However, several challenges persist. Usually, it is difficult to design a reward function to achieve good training results. At the same time, a large number of training samples are required to achieve a certain level of performance for the model [14]. It makes it necessary to debug the code for a long time to implement the strategy of obtaining models through reinforcement learning training, and the time cost of training is not significant. Furthermore, compared to classical methods, they lack interpretability [15]. After a lot of training, some detailed problems such as not being able to overcome certain obstacles cannot be explained very accurately, which makes it very difficult to complete debugging and improvement work, and it is also not easy to apply the successfully trained model very appropriately to practical scenarios.

III. THE PROBLEM OF QUADRUPED ROBOT WALKING

A. Problem Statement

The virtual environment for Spot to perform search tasks is a mine tunnel, as shown in Figure 1. The tunnel is relatively spacious, with an intersection at the end. The tunnel is not capped and is surrounded by walls with a height of 15 meters on all four sides. The tunnel has a width of 10 and a length of 17 meters. It is expected that during the environment generation process, the victim will be represented by a spherical object and appear randomly during the environment generation process. The robot needs to search for the victim without touching the wall and then approach the victim. However, since this article does not achieve long-distance movement of the robot, this setting is aimed at further research in the future. The strategy of the agent in this task can be described as follows: achieving posture correction learning through self-state observation and ultimately learning to walk.

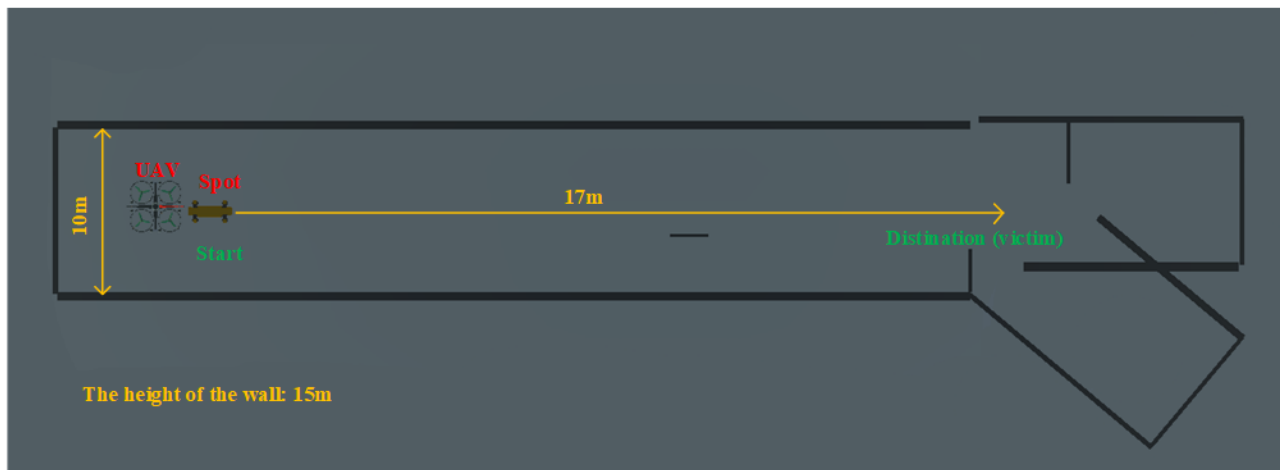


Fig. 1. Virtual scene with a quadruped robot Spot and a UAV.

The goal is to train agents through reinforcement learning algorithms and identify the optimal policy network.

B. Reinforcement Learning

In this work, quadruped robot is given approximate range limits for each joint angle, and the joint motion speed is regulated through StepCount division and setting joint parameters. Regarding UAV, this study directly adopted mature training methods for UAV target finding. The brief training process is shown in Figure 2.

1) *Action space*: Although in real life, the control of robots should be continuous. During the training process, in order to prevent quadruped robot from falling into local optimal states, this paper discretizes the actions of Spot. In the simulation environment, robot need to continuously penetrate deep into mines to complete search and rescue tasks. At this point, only forward movement is considered effective action. Therefore, discretizing the action space helps to improve training efficiency. Spot has twelve available discrete actions, consisting of two types of planes and two types of joints. In order to reduce the complexity of this system, one planar type was removed, and eight action spaces were used in this paper.

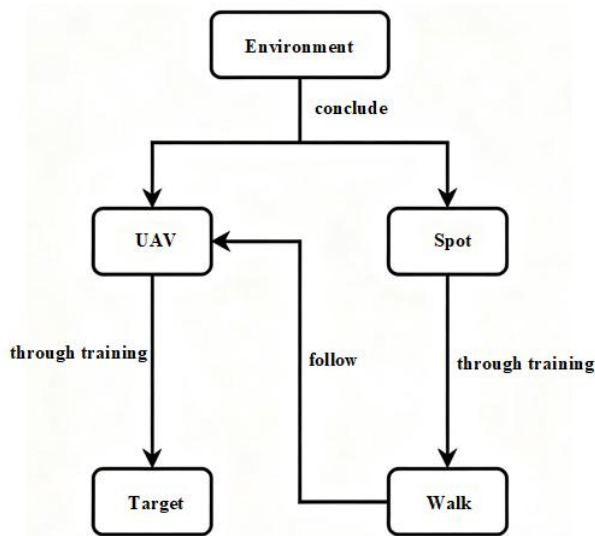


Fig. 2. Training process.

The discretized action space is shown in Table 1. The remaining plane is the x-y plane shown in Part IV below.

TABLE I. DISCRETIZED ACTION SPACE FOR SPOT (- FOR LEFT / + FOR RIGHT)

Discrete Action ID	Angle ratio	Basic Direction
1	0.28	0.989
2	0.50	-1.885
4	0.28	0.989
5	0.50	-1.885
7	0.28	0.989
8	0.50	-1.880
10	0.28	0.989
11	0.50	-1.880

In future further training, to prevent excessive discretization from limiting the flexibility of robots in real-world environments, a gradual transition strategy can be adopted. Using coarse-grained discrete action spaces in the early stages of training to help strategies quickly master basic motor skills. As the training progresses, the action space is gradually refined, the degree of freedom of actions is increased, and the strategy gradually adapts to a more continuous control mode. In addition, hierarchical training can also be considered, starting with training low complexity subtasks and then expanding to high degree of freedom control.

2) *Observation Space*: In the decision-making process, Spot needs to perform the following behaviors based on their relative positions and postures. Spot can obtain a relative position vector with the expected distance, which the robot can use to determine the following strategies in the next step. In addition, Spot can also obtain its own self positioning, speed, direction, and rotation angle in three-dimensional space, thus enabling precise posture control.

Afterwards, by measuring and observing the rotation angle of each axis, i.e. Euler Angles, and setting rewards, the robot is motivated to continuously adjust and learn towards the correct posture, achieving standing, limb motion allocation, posture adjustment, and ultimately walking. Therefore, inspired by the concept of course learning, this article adopts a curiosity-based reinforcement learning method to promote robot learning and has achieved certain results.

IV. EXPERIMENT AND RESULTS

As shown in Figure 3, a quadruped robot Spot is included in the simulation environment. In each episode, the robot will reset its position. Among them, the annotation number is the joint action space number. 2, 5, 8, and 11 respectively represent the motion space of the robot's left front, right front, left rear, and right rear knee parts. 1, 4, 7, and 10 respectively represent the motion space of the left front, right front, left rear, and right rear hip parts of the robot. These joint movements mentioned above only belong to the x-y plane. 0, 3, 6, and 9 represent the motion spaces of the left front, right front, left rear, and right rear hip parts of the robot, respectively. However, they represent the joint movements in the y-z plane. In order to avoid overly redundant and complex control of the motion space, this paper fixes these four motion spaces and does not include them as part of the training. This means that during training, the joints of the robotic dog only rotate in the X-Y plane.

A. Parameter Settings

Table 2 details the hyperparameters of the PPO2-ICM algorithm as well as the rewards used. Training tool versions are as follows: mlagents-toolkits: 0.30.0; ml-agents-envs: 0.30.0; Communicator API: 1.5.0; PyTorch: 2.4.0+cu118. The training process was carried out on an 12th Gen Intel(R) Core(TM) i7-12700H CPU and a NVIDIA GeForce RTX 3070 Ti Laptop GPU. The Unity version used for training is 2022.3.4f1c1. In numerous example environments, multiple copies of the training area are created within a single scene, which typically accelerates the training process by enabling parallel experience gathering [16]. UAV is trained using this method, as shown in Figure 4. However, due to the lack of stable parameters obtained by Spot, multiple instances will not run simultaneously in the same scene. So the two robots are temporarily trained separately, but in the same environment. After obtaining more stable parameters in the future, a multi process approach can be adopted and achieve collaborative training of two types of robots. Figure 5 respectively shows the Cumulative Reward curve and Episode Length curve during the partial training process.

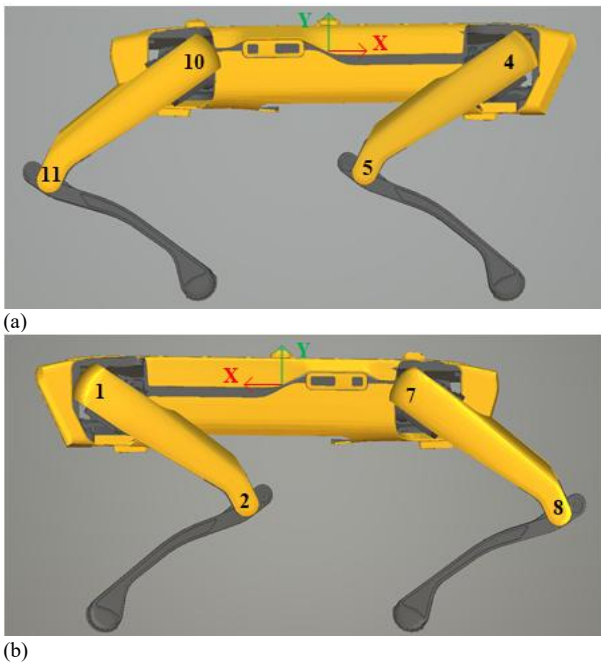


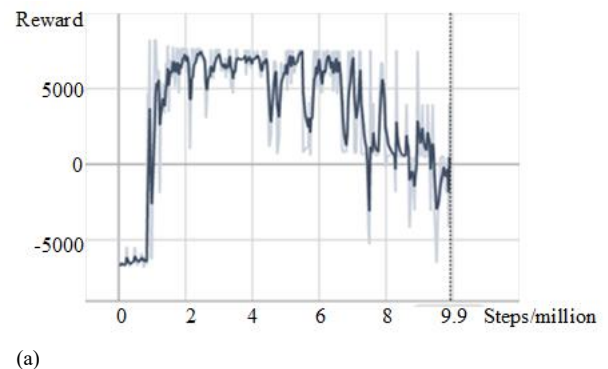
Fig. 3. (a) Spot right side view. (b) Spot left side view.

TABLE II. HYPERPARAMETERS USED IN SPOT TRAINING PROCESS

Parameter	Spot	Reward	Punishment
Learning rate	0.0003	$R_{\text{forward}}=0.6$	
X-axis deflection		$R_x = 0.6f / (1f + \text{tiltAngle}_{abx})$	$P_x = -0.1$
Y-axis deflection		$R_y = 0.6f / (1f + \text{tiltAngle}_{aby})$	$P_y = -0.1$
Z-axis deflection		$R_z = 0.6f / (1f + \text{tiltAngle}_{abz})$	$P_z = -0.1$
Advance coefficient		$0.6f * (1 - (5 - \text{localPosition.x}) / 20f)$	
Invalid steps			-0.05
Learning rate schedule	Linear		
Extrinsic gamma	0.99		
Extrinsic strength	1.0		
Curiosity gamma	0.99		
Curiosity strength	0.02		
Curiosity learning rate	0.0003		

In Figure 4, the reward curve in the image above rapidly increases from around -7000 to around 7000 during the 0-2M steps, indicating that the drone quickly learned the basic strategy. During the 2M to 6M steps, the reward remains at a high level, indicating that the training is relatively stable and the strategy tends to converge. After 6M to 10M steps, the reward begins to decrease and fluctuate, indicating that the model has overfitting. Therefore, training with less than 6M steps is appropriate for UAV. In the Episode Length curve in the image below, the UAV has the most stable training duration per round around 2M-4M, indicating the best training effect during this period.

In Figure 5, the reward curve in the image above tends to stabilize after experiencing fluctuations of 400K steps, indicating that Spot is relatively stable in maintaining balance and moving forward tasks thereafter. However, from a numerical perspective, the overall reward has increased from around -400 to around 40, indicating that the overall reward obtained is not much and the current task setting is difficult. From the Episode Length chart in the image below, it can be seen that the overall fluctuation is not significant, maintaining a position around 200. The duration of each training round is relatively stable, and the degree of achieving the goal is stable.



(a)

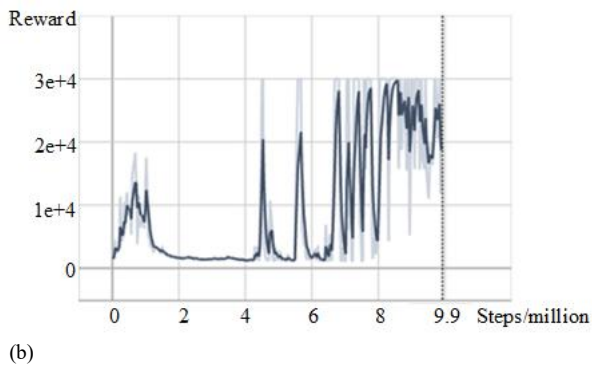


Fig. 4. Cumulative Reward curve (a) and Episode Length (b) of UAV.

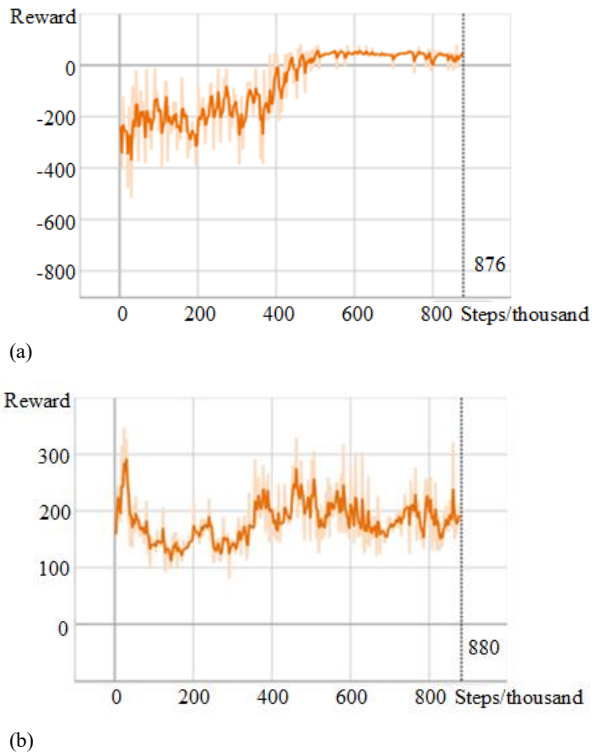


Fig. 5. Cumulative Reward curve (a) and Episode Length (b) of Spot

B. Inference Experiment

In order to evaluate the effectiveness of the proposed model in performing navigation tasks in heterogeneous robot systems, inference testing was conducted in a simulated environment. In inference mode, the behavior of Spot is determined by its corresponding policy network. In the training environment, Spot moves from the starting point. During the reasoning process, the quadruped robot Spot can exhibit a movement posture that tends towards normal and can advance a certain distance about 1. This indicates that although the movement over a long distance was not achieved in the training results, its posture demonstrated a certain degree of robustness during the training process. For multi joint quadruped robot with up to 12 motion spaces, the algorithm still demonstrates its effectiveness, but more comprehensive content and more work support are clearly needed in terms of walking. This experiment successfully validated the feasibility of the proposed approach and accumulated rich experience for future work. Based on this research, reward shaping can be conducted in the future to improve overall performance and adaptability, or the model

can be tested on real robots, for obtaining more stable strategies and results.

V. CONCLUSION

This article proposes a quadruped robot motion control strategy based on curiosity guided reinforcement learning (CGRL). This strategy applies a reinforcement learning training framework to initiate the training of individual agents based on specific tasks. To address the issue of sparse rewards in the environment, the PPO2 algorithm and Intrinsic Curiosity Module (ICM) were employed to provide agents with intrinsic exploration rewards. The experimental results indicate that the quadruped robot (Spot) is capable of performing various complex movements such as leg lifting, laying a preliminary foundation for challenging more complex environments in the future. The four point contact walking mode can be achieved. Tests in simulated environments have shown that quadruped robots can optimize their posture based on the training process and successfully achieve short distance advancement. This indicates that after further optimization, quadruped robots can complete navigation tasks in completely unknown environments. This method and results have important practical significance for the research of future collaborative robot systems.

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